

Given a map $f: \mathbb{R} \rightarrow \mathbb{R}$, a map

$$g: x \mapsto \int_J k(x, t) f(t) dt$$

is called the 'integral transform'.

In this page, we cover typical int. forms.

Laplace transform.

Given a map $f: \mathbb{R} \rightarrow \mathbb{C}$, a Laplace transform of f is given by:

$$\mathcal{L}f: \mathbb{C} \rightarrow \mathbb{C}; x \mapsto \int_0^{\infty} e^{-xt} \cdot f(t) dt.$$

Prop. If f is differentiable and $e^{-xR} \cdot f(R) \rightarrow 0$ as $R \rightarrow \infty$,

then $(\mathcal{L}f)(x) = \int_0^{\infty} e^{-xt} \cdot f(t) dt = \lim_{R \rightarrow \infty} \left[e^{-xt} f(t) \Big|_0^R \right] + x \cdot \int_0^{\infty} e^{-xt} \cdot f(t) dt$

$$= x \cdot (\mathcal{L}f)(x) - f(0)$$

Fourier transform.

